Jacob Yankee
COSC 461
Maniccam

The parameters used to obtain a minimum 50% accuracy are:

- 12 epochs
- 2 hidden layers
- 200 size hidden layers
- Sgd optimizer

```python
import tensorflow as tf

data = tf.keras.datasets.cifar10

(training_images, training_labels), (test_images, test_labels) = data.load_data()

training_images = training_images/255
test_images = test_images/255

model = tf.keras.models.Sequential([
            tf.keras.layers.Flatten(input_shape=(32, 32, 3)),
            tf.keras.layers.Dense(200, activation=tf.nn.relu),
            tf.keras.layers.Dense(200, activation=tf.nn.relu),
            tf.keras.layers.Dense(10, activation=tf.nn.softmax)
        ])

model.compile(optimizer='sgd',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

model.fit(training_images, training_labels, epochs=12)

model.evaluate(test_images, test_labels)

classifications = model.predict(test_images)

print(classifications[0])
print(test_labels[0])
```

```
Epoch 1/12
1563/1563 [==============================] - 11s 7ms/step - loss: 1.8830 - accuracy: 0.3258
Epoch 2/12
1563/1563 [==============================] - 11s 7ms/step - loss: 1.6945 - accuracy: 0.4000
Epoch 3/12
1563/1563 [==============================] - 10s 7ms/step - loss: 1.6139 - accuracy: 0.4294
Epoch 4/12
1563/1563 [==============================] - 13s 9ms/step - loss: 1.5622 - accuracy: 0.4462
Epoch 5/12
1563/1563 [==============================] - 10s 7ms/step - loss: 1.5182 - accuracy: 0.4623
Epoch 6/12
1563/1563 [==============================] - 11s 7ms/step - loss: 1.4807 - accuracy: 0.4747
Epoch 7/12
1563/1563 [==============================] - 11s 7ms/step - loss: 1.4503 - accuracy: 0.4875
Epoch 8/12
1563/1563 [==============================] - 11s 7ms/step - loss: 1.4209 - accuracy: 0.4970
Epoch 9/12
1563/1563 [==============================] - 11s 7ms/step - loss: 1.3962 - accuracy: 0.5054
Epoch 10/12
1563/1563 [==============================] - 12s 8ms/step - loss: 1.3717 - accuracy: 0.5136
Epoch 11/12
1563/1563 [==============================] - 10s 7ms/step - loss: 1.3505 - accuracy: 0.5225
Epoch 12/12
1563/1563 [==============================] - 11s 7ms/step - loss: 1.3272 - accuracy: 0.5286
313/313 [==============================] - 1s 4ms/step - loss: 1.3876 - accuracy: 0.5020
313/313 [==============================] - 1s 4ms/step
[0.03965699 0.01475109 0.1096591  0.4145318  0.06624536 0.18232526
 0.08151652 0.00334941 0.08620492 0.00175963]
[3]
```